

CLIMATIC EYE

Hardware Connection & Power Integration Guide

ESP32-based environmental monitoring node

SYSTEM	CONFIGURATION
Controller	Generic 30-pin ESP32 DevKit
Environmental sensing	BMP280, BH1750, DHT11
Field sensing	Raindrop sensor, soil-moisture sensor
Positioning	NEO-6M GPS
Wireless link	RA-02 LoRa
Power	Solar panels, TP4056, Li-ion battery, 5 V boost converter

Document purpose
Use this guide as the wiring reference during assembly and bench testing. The GPIO allocation is kept consistent with the Climatic Eye firmware and admin debug dashboard.

Revision: Prototype Integration Guide | Bench assembly and field-node development

1. System Overview

The node combines environmental sensors, GPS, LoRa communications, and a solar-charged battery supply around a generic 30-pin ESP32. BMP280 and BH1750 share the I²C bus. The NEO-6M uses a dedicated hardware UART. The RA-02 uses the ESP32 VSPI pins. The raindrop and soil sensors use ADC inputs.

Functional architecture

Solar panels → TP4056 → Li-ion battery → 5 V boost converter → ESP32 VIN/5V → ESP32 3.3 V rail → sensors + RA-02

Core rule

Keep all grounds common. Keep the RA-02 on 3.3 V only. Verify the boost-converter output with a multimeter before connecting it to the ESP32.

2. Complete GPIO Assignment

Device	Signal	ESP32 GPIO	Interface / note
BMP280	SDA	GPIO 21	I ² C
BMP280	SCL	GPIO 22	I ² C
BH1750	SDA	GPIO 21	Shares I ² C with BMP280
BH1750	SCL	GPIO 22	Shares I ² C with BMP280
NEO-6M	TX	GPIO 16	ESP32 UART2 RX
NEO-6M	RX	GPIO 17	ESP32 UART2 TX
DHT11	DATA	GPIO 27	Digital input
Raindrop module	AO	GPIO 34	ADC input-only
Raindrop module	DO	Not connected	Analog reading used
Soil moisture module	AO	GPIO 35	ADC input-only
Soil moisture module	DO	Not connected	Analog reading used
RA-02	SCK	GPIO 18	VSPI
RA-02	MISO	GPIO 19	VSPI
RA-02	MOSI	GPIO 23	VSPI
RA-02	NSS / CS	GPIO 5	Chip select
RA-02	RESET	GPIO 14	Reset control
RA-02	DIO0	GPIO 26	LoRa interrupt

3. Sensor Connections

3.1 BMP280

BMP280 pin	ESP32 connection	Instruction
VCC / VIN	3.3 V	Use 3.3 V.
GND	GND	Common ground.
SDA	GPIO 21	I ² C data.
SCL	GPIO 22	I ² C clock.

3.2 BH1750

BH1750 pin	ESP32 connection	Instruction
VCC	3.3 V	Use 3.3 V.
GND	GND	Common ground.
SDA	GPIO 21	Shares SDA with BMP280.
SCL	GPIO 22	Shares SCL with BMP280.

I²C bus

BMP280 and BH1750 are connected in parallel on GPIO 21/22. If either is not detected, check power, ground, SDA/SCL orientation, and run an I²C scanner.

3.3 DHT11

DHT11 pin	ESP32 connection	Instruction
VCC	3.3 V	Use 3.3 V.
GND	GND	Common ground.
DATA	GPIO 27	Digital data line.

For a bare 4-pin DHT11, add approximately a 10 k Ω pull-up resistor between DATA and 3.3 V. Common 3-pin DHT11 modules normally include the pull-up resistor.

3.4 Raindrop sensor module

Module pin	ESP32 connection	Instruction
VCC	3.3 V	Prefer 3.3 V so AO stays within the ESP32 ADC range.
GND	GND	Common ground.
AO	GPIO 34	Analog measurement.
DO	Not connected	Not used by current firmware.

3.5 Soil-moisture sensor

Module pin	ESP32 connection	Instruction
VCC	3.3 V	Prefer 3.3 V for ADC compatibility.
GND	GND	Common ground.

Module pin	ESP32 connection	Instruction
AO	GPIO 35	Analog measurement.
DO	Not connected	Not used by current firmware.

4. GPS and LoRa Connections

4.1 NEO-6M GPS

NEO-6M pin	ESP32 connection	Instruction
VCC	5 V/VIN or module-specified supply	Verify the exact breakout board.
GND	GND	Common ground.
TX	GPIO 16	GPS TX → ESP32 UART2 RX.
RX	GPIO 17	ESP32 UART2 TX → GPS RX.

GPS fix

Initial position acquisition can take time. Test outdoors or near a window with a clear sky view. A valid module can output NMEA data before obtaining a position fix.

4.2 RA-02 LoRa

RA-02 pin	ESP32 connection	Instruction
VCC	3.3 V ONLY	Never connect to 5 V.
GND	GND	Common ground.
SCK	GPIO 18	VSPI clock.
MISO	GPIO 19	VSPI MISO.
MOSI	GPIO 23	VSPI MOSI.
NSS / CS	GPIO 5	Chip select.
RESET	GPIO 14	Reset control.
DIO0	GPIO 26	Required by current firmware.
DIO1-DIO5	Not connected	Not required by current firmware.

RA-02 protection

Use a stable 3.3 V supply. Keep the antenna connected whenever transmitting. A local 100 μ F capacitor across RA-02 3.3 V and GND is recommended to reduce supply dips during TX.

LoRa frequency

The firmware currently uses 433E6 as the default frequency. Change it only if your actual RA-02 hardware and legal operating band require another frequency. The receiver must use the same frequency and compatible radio settings.

5. Solar, Charging, Battery and Boost Connections

The prototype power path uses a single-cell Li-ion battery and a 5 V boost converter. The TP4056 is a single-cell Li-ion charger, so the battery arrangement must remain a 1-cell nominal system.

5.1 Solar panels → TP4056

Solar connection	TP4056	Instruction
Panel 1 +	IN+	Positive input.
Panel 1 –	IN–	Negative input.
Panel 2 +	IN+	Parallel positive connection.
Panel 2 –	IN–	Parallel negative connection.

The two panels are intended to be connected in parallel for the prototype. Do not connect them in series unless the resulting voltage has been verified as suitable for the specific charger.

5.2 TP4056 → battery

TP4056	Battery / load	Instruction
B+	Battery +	Battery positive.
B–	Battery –	Battery negative.
OUT+	Booster IN+	Load output positive.
OUT–	Booster IN–	Load output negative.

Two-cell warning

Do not connect two 3.7 V cells in series to a standard single-cell TP4056. If two compatible cells are used with this charger, they must form a properly matched parallel 1S2P pack, with both cells at essentially the same voltage before paralleling.

5.3 Booster → ESP32

Booster	ESP32	Instruction
OUT+	VIN / 5V	Feed approximately 5.0 V.
OUT–	GND	Common ground.

Mandatory pre-power check

Before connecting the booster to the ESP32, measure its output with a multimeter and set it to approximately 5.0 V. Do not connect an unknown or unadjusted boost output to the ESP32 VIN/5V input.

5.4 ESP32 3.3 V distribution

Use the ESP32 3.3 V rail for BMP280, BH1750, DHT11, raindrop module, soil-moisture module, and RA-02 as specified in this guide.

6. Power Architecture Reference

Solar Panel 1 + ■■■
 Solar Panel 2 + ■■■■ TP4056 IN+
 Solar Panel 1 - ■■
 Solar Panel 2 - ■■■■ TP4056 IN-

 TP4056 B+ ■■■■■■■■ Battery +
 TP4056 B- ■■■■■■■■ Battery -

 TP4056 OUT+ ■■■■■■■■ Booster IN+
 TP4056 OUT- ■■■■■■■■ Booster IN-

 Booster OUT+ ■■■■■■ ESP32 VIN / 5V
 Booster OUT- ■■■■■■ ESP32 GND

 ESP32 3.3V ■■■■■■■■ Sensors + RA-02
 ESP32 GND ■■■■■■■■ All grounds

7. Assembly and Power-Up Procedure

Step	Action
1	Inspect every module and confirm its pin labels before wiring.
2	Verify solar-panel polarity with a multimeter.
3	Connect the solar panels to TP4056 IN+ / IN- in parallel.
4	Connect the Li-ion battery to TP4056 B+ / B- using a suitable single-cell configuration.
5	Connect TP4056 OUT+ / OUT- to the booster input.
6	Power the booster and adjust its output to approximately 5.0 V.
7	With power removed, connect booster output to ESP32 VIN/5V and GND.
8	Connect the 3.3 V sensor rail and common grounds.
9	Connect I ² C sensors, DHT11, analog sensors, and GPS.
10	Connect the RA-02 last. Confirm 3.3 V supply and attach the antenna.
11	Power the node and check Serial Monitor at 115200 baud.
12	Verify sensor initialization and LoRa status before field use.

8. Verification Checklist

Check	Expected result
ESP32 supply	Approximately 5.0 V at VIN/5V input.
ESP32 3.3 V rail	Stable 3.3 V output.
BMP280	Detected at I ² C address 0x76 or 0x77.
BH1750	Detected and produces non-negative lux readings.
DHT11	Temperature and humidity readings are valid.
Raindrop	GPIO 34 ADC value changes when the plate becomes wet.
Soil sensor	GPIO 35 ADC value changes between dry and wet conditions.

Check	Expected result
GPS	NMEA data appears; valid fix may require outdoor sky view.
RA-02	Initializes and transmits with antenna attached.
Web debug page	ESP32 hotspot or configured Wi-Fi provides access.

9. Important Notes, Exceptions and Fallbacks

Sensor failure handling

Unavailable readings must remain unavailable rather than being replaced by fabricated values. The admin debug page uses N/A when a reading cannot be obtained. A missing GPS fix is represented separately from a GPS hardware failure.

Analog sensor calibration

Rain and soil percentages depend on calibration. Record actual raw ADC values under known dry and wet conditions, then update the firmware calibration constants. Retain raw ADC values for diagnostics.

Power fallback

If the solar supply is weak, the TP4056 may not charge effectively. For final outdoor deployment, use a charger/power-path solution designed for solar input rather than relying on a conventional TP4056 as the complete solar power-management system.

Common faults

Symptom	First checks
ESP32 resets during LoRa TX	RA-02 supply stability, common ground, local 100 μ F capacitor, antenna, booster capacity.
BMP280/BH1750 not detected	3.3 V, GND, SDA GPIO 21, SCL GPIO 22, I ² C address.
DHT11 errors	DATA GPIO 27, pull-up for bare sensor, timing, 3.3 V supply.
Rain/soil percentage wrong	Check raw ADC first; recalibrate dry/wet endpoints.
GPS has no fix	Allow acquisition time; use outdoor/clear-sky position.
LoRa initialization fails	3.3 V, CS GPIO 5, RESET GPIO 14, DIO0 GPIO 26, SPI pins, frequency.
ESP32 will not boot	Disconnect peripherals, verify power rails, reconnect modules one at a time.
Boost output too high	Disconnect ESP32 immediately; adjust booster using a multimeter.

Design constraints

Constraint	Rule
RA-02 supply	3.3 V only.
GPIO 34 / 35	Input-only; use for ADC sensors.
GPIO 6-11	Do not use; associated with ESP32 flash on typical modules.
TP4056	Single-cell Li-ion charging architecture.
Battery paralleling	Only suitable matched cells at essentially equal voltage.
USB + external supply	Avoid simultaneous USB and external VIN/5V unless the exact board power path is verified.

Final pre-deployment check

Run the complete node from the intended battery and booster for an extended bench test. Confirm sensor readings, LoRa transmission, GPS acquisition, ESP32 stability, charging behavior, and enclosure ventilation before field installation.

10. Firmware and Debug Interface

The current debug firmware exposes structured telemetry suitable for later dashboard integration. Principal fields include `node_id`, `packet`, `temperature`, `humidity`, `pressure`, `altitude`, `light_lux`, `rain_raw`, `rain_percent`, `soil_raw`, `soil_percent`, `gps_fix`, `latitude`, `longitude`, `satellites`, sensor status flags, and LoRa transmission status.

The ESP32 admin debug page is a temporary local diagnostic interface. When no router credentials are configured, the ESP32 creates the ClimaticEye-Debug access point and serves the page at its local access-point address.